11.2 Goodness of fit tests when all parameters are specified

Suppose that n independent random variables — $Y_1, \ldots, Y_n$, each taking on one of the values $1, 2, \ldots, k$ — are to be observed and we are interested in testing the null hypothesis that $\{p_i, i = 1, \ldots, k\}$ is the probability mass function of the Y_j. That is, if Y represents any of the Y_j, then the null hypothesis is

$$H_0 : P\{Y = i\} = p_i, \qquad i = 1, \ldots, k$$

whereas the alternative hypothesis is

$$H_1 : P\{Y = i\} \neq p_i, \qquad \text{for some } i = 1, \ldots, k$$

To test the foregoing hypothesis, let $X_i, i = 1, \ldots, k$, denote the number of the Y_j's that equal i. Then as each Y_j will independently equal i with probability $P\{Y = i\}$, it follows that, under H_0, X_i is binomial with parameters n and p_i. Hence, when H_0 is true,

$$E[X_i] = np_i$$

and so $(X_i - np_i)^2$ will be an indication as to how likely it appears that p_i indeed equals the probability that $Y = i$. When this is large, say, in relationship to np_i, then it is an indication that H_0 is not correct. Indeed such reasoning leads us to consider the following test statistic:

$$T = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \tag{11.2.1}$$

and to reject the null hypothesis when T is large.

To determine the critical region, we need first specify a significance level α and then we must determine that critical value c such that

$$P_{H_0}\{T \geq c\} = \alpha$$

That is, we need to determine c so that the probability that the test statistic T is at least as large as c, when H_0 is true, is α. The test is then to reject the hypothesis, at the α level of significance, when $T \geq c$ and to accept when $T < c$.

It remains to determine c. The classical approach to doing so is to use the result that when n is large T will have, when H_0 is true, approximately (with the approximation becoming exact as n approaches infinity) a chi-square distribution with $k - 1$ degrees of freedom. Hence, for n large, c can be taken to equal $\chi^2_{\alpha,k-1}$; and so the approximate α level test is

reject H_0 if $T \geq \chi^2_{\alpha,k-1}$

accept H_0 otherwise

If the observed value of T is $T = t$, then the preceding test is equivalent to rejecting H_0 if the significance level α is at least as large as the p-value given by

$$\begin{aligned} p\text{-value} &= P_{H_0}\{T \geq t\} \\ &\approx P\{\chi^2_{k-1} \geq t\} \end{aligned}$$

where χ^2_{k-1} is a chi-square random variable with $k-1$ degrees of freedom.

An accepted rule of thumb as to how large n need be for the foregoing to be a good approximation is that it should be large enough so that $np_i \geq 1$ for each $i, i = 1, \ldots, k$, and also at least 80 percent of the values np_i should exceed 5.

Remarks

(a) A computationally simpler formula for T can be obtained by expanding the square in Equation (11.2.1) and using the results that $\sum_i p_i = 1$ and $\sum_i X_i = n$ (why is this true?):

$$\begin{aligned} T &= \sum_{i=1}^{k} \frac{X_i^2 - 2np_iX_i + n^2p_i^2}{np_i} \qquad (11.2.2) \\ &= \sum_i X_i^2/np_i - 2\sum_i X_i + n\sum_i p_i \\ &= \sum_i X_i^2/np_i - n \end{aligned}$$

(b) The intuitive reason why T, which depends on the k values $X_1, \ldots, X_k$, has only $k-1$ degrees of freedom is that 1 degree of freedom is lost because of the linear relationship $\sum_i X_i = n$.

(c) Whereas the proof that, asymptotically, T has a chi-square distribution is advanced, it can be easily shown when $k = 2$. In this case, since $X_1 + X_2 = n$, and $p_1 + p_2 = 1$, we see that

$$\begin{aligned} T &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(n - X_1 - n[1-p_1])^2}{n(1-p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1-p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1(1-p_1)} \qquad \text{since} \qquad \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)} \end{aligned}$$

However, X_1 is a binomial random variable with mean np_1 and variance $np_1(1-p_1)$ and thus, by the normal approximation to the binomial, it follows

that $(X_1 - np_1)/\sqrt{np_1(1-p_1)}$ has, for large n, approximately a standard normal distribution, and so its square has approximately a chi-square distribution with 1 degree of freedom.

Example 11.2.a. In recent years, a correlation between mental and physical well-being has increasingly become accepted. An analysis of birthdays and death days of famous people could be used as further evidence in the study of this correlation. To use these data, we are supposing that being able to look forward to something betters a person's mental state, and that a famous person would probably look forward to his or her birthday because of the resulting attention, affection, and so on. If a famous person is in poor health and dying, then perhaps anticipating his birthday would "cheer him up and therefore improve his health and possibly decrease the chance that he will die shortly before his birthday." The data might therefore reveal that a famous person is less likely to die in the months before his or her birthday and more likely to die in the months afterward.

Solution. To test this, a sample of 1251 (deceased) Americans was randomly chosen from *Who Was Who in America*, and their birth and death days were noted. (The data are taken from D. Phillips, "Death Day and Birthday: An Unexpected Connection," in *Statistics: A Guide to the Unknown*, Holden-Day, 1972.) The data are summarized in Table 11.1.

If the death day does not depend on the birthday, then it would seem that each of the 1251 individuals would be equally likely to fall in any of the 12 categories. Thus, let us test the null hypothesis

$$H_0 = p_i = \frac{1}{12}, \qquad i = 1, \ldots, 12$$

Since $np_i = 1251/12 = 104.25$, the chi-square test statistic for this hypothesis is

$$T = \frac{(90)^2 + (100)^2 + (87)^2 + \cdots + (106)^2}{104.25} - 1251$$
$$= 17.192$$

The p-value is

$$\begin{aligned} p\text{-value} &\approx P(\chi^2_{11} \geq 17.192) \\ &> 1 - \text{pchisq}(17.192, 11) \\ &[1] \quad 0.1023232 \end{aligned}$$

The results of this test leave us somewhat up in the air about the hypothesis that an approaching birthday has no effect on an individual's remaining lifetime. For whereas the data are not quite strong enough (at least, at

Table 11.1 Number of Deaths Before, During, and After the Birth Month.

	6 Months Before	5 Months Before	4 Months Before	3 Months Before	2 Months Before	1 Month Before	The Month	1 Month After	2 Months After	3 Months After	4 Months After	5 Months After
Number of deaths	90	100	87	96	101	86	119	118	121	114	113	106

$n = 1251$.
$n/12 = 104.25$.

the 10 percent level of significance) to reject this hypothesis, they are certainly suggestive of its possible falsity. This raises the possibility that perhaps we should not have allowed as many as 12 data categories, and that we might have obtained a more powerful test by allowing for a fewer number of possible outcomes. For instance, let us determine what the result would have been if we had coded the data into 4 possible outcomes as follows:

outcome $1 = -6, -5, -4$
outcome $2 = -3, -2, -1$
outcome $3 = 0, 1, 2$
outcome $4 = 3, 4, 5$

That is, for instance, an individual whose death day occurred 3 months before his or her birthday would be placed in outcome 2. With this classification, the data would be as follows:

Outcome	Number of Times Occurring
1	277
2	283
3	358
4	333

$n = 1251$.
$n/4 = 312.75$.

The test statistic for testing $H_0 = p_i = 1/4, i = 1, 2, 3, 4$ is

$$T = \frac{(277)^2 + (283)^2 + (358)^2 + (333)^2}{312.75} - 1.251$$
$$= 14.775$$

Hence, as $\chi^2_{.01,3} = 11.345$, the null hypothesis would be rejected even at the 1 percent level of significance. Indeed, R yields

$$p\text{-value} = P\{\chi^2_3 \geq 14.775\} = 1 - \text{pchisq}(14.775, 3) = 0.00201938$$

The foregoing analysis is, however, subject to the criticism that the null hypothesis was chosen after the data were observed. Indeed, while there is nothing incorrect about using a set of data to determine the "correct way" of phrasing a null hypothesis, the additional use of those data to test that very hypothesis is certainly questionable. Therefore, to be quite certain of the conclusion to be drawn from this example, it seems prudent to choose a second random sample — coding the values as before — and again test $H_0 : p_i = 1/4, i = 1, 2, 3, 4$ (see Problem 3). ■

R can be used to both compute the value of the test statistic and determine the resulting p-value. Suppose $x = c(x_1, \ldots, x_k)$ is the vector giving the number of times that each of the k possible values occurred, and suppose we want to test the null hypothesis that the probability vector for outcomes is $p = c(p_1, \ldots, p_k)$. To do so, just use the R commands

$$\begin{aligned} &> \quad x = c(x_1, \ldots, x_k) \\ &> \quad \text{chisq.test}(x,\ p = c(p_1, \ldots, p_k)) \end{aligned}$$

For instance, for the example just considered

$$\begin{aligned} &> \qquad x = c(277, 283, 358, 333) \\ &> \qquad \text{chisq.test}(x,\ p = c(1/4, 1/4, 1/4, 1/4)) \\ &\qquad\quad \text{Chi-squared test for given probabilities} \\ &\text{data :} \qquad x \\ &X - \text{squared} = 14.775, \quad \text{df} = 3, \quad p\text{-value} = 0.00202 \end{aligned}$$

Example 11.2.b. A contractor who purchases a large number of fluorescent lightbulbs has been told by the manufacturer that these bulbs are not of uniform quality but rather have been produced in such a way that each bulb produced will, independently, either be of quality level A, B, C, D, or E, with respective probabilities .15, .25, .35, .20, .05. However, the contractor feels that he is receiving too many type E (the lowest quality) bulbs, and so he decides to test the producer's claim by taking the time and expense to ascertain the quality of 30 such bulbs. Suppose that he discovers that of the 30 bulbs, 3 are of quality level A, 6 are of quality level B, 9 are of quality level C, 7 are of quality level D, and 5 are of quality level E. Do these data, at the 5 percent level of significance, enable the contractor to reject the producer's claim?

Solution. We use R as follows:

$$\begin{aligned} &> \qquad x = c(3, 6, 9, 7, 5) \\ &> \qquad \text{chisq.test}(x,\ p = c(.15, .25, .35, .20, .05)) \\ &\qquad\quad \text{Chi-squared test for given probabilities} \\ &\text{data : } x \\ &X - \text{squared} = 9.3476, \quad \text{df} = 4, \quad \text{p-value} = 0.05297 \end{aligned}$$

$$\begin{aligned} &\text{Warning message:} \\ &\text{In chi-sq.test}(x,\ p = c(.15, .25, .35, .20, .05)) \\ &\text{Chi-squared approximation may be incorrect} \end{aligned}$$

Note that R gives the warning that, because of the relatively small numbers falling in each category, the chi-squared approximation for the distribution of the test statistic may not be totally accurate. Assuming that its accuracy is

not an issue, the hypothesis would not be rejected at the 5 percent level of significance (but since it would be rejected at all significance levels above .053, the contractor should certainly remain skeptical). ■

11.2.1 Determining the critical region by simulation[1]

From 1900 when Karl Pearson first showed that T has approximately (becoming exact as n approaches infinity) a chi-square distribution with $k-1$ degrees of freedom, until relatively recently, this approximation was the only means available for determining the p-value of the goodness of fit test. However, with the advent of inexpensive, fast, and easily available computational power a second, potentially more accurate, approach has become available: namely, the use of simulation to obtain to a high level of accuracy the p-value of the test statistic.

The simulation approach is as follows. First, the value of T is determined — say, $T = t$. Now to determine whether or not to accept H_0, at a given significance level α, we need to know the probability that T would be at least as large as t when H_0 is true. To determine this probability, we simulate n independent random variables $Y_1^{(1)}, \ldots, Y_n^{(1)}$ each having the probability mass function $\{p_i, i = 1, \ldots, k\}$ — that is,

$$P\{Y_j^{(1)} = i\} = p_i, \qquad i = 1, \ldots, k, \qquad j = 1, \ldots, n$$

Now let

$$X_i^{(1)} = \text{number } j : Y_j^{(1)} = i$$

and set

$$T^{(1)} = \sum_{i=1}^{k} \frac{(X_i^{(1)} - np_i)^2}{np_i}$$

Now repeat this procedure by simulating a second set, independent of the first set, of n independent random variables $Y_1^{(2)}, \ldots, Y_n^{(2)}$ each having the probability mass function $\{p_i, i = 1, \ldots, k\}$ and then, as for the first set, determining $T^{(2)}$. Repeating this a large number, say, r, of times yields r independent random variables $T^{(1)}, T^{(2)}, \ldots, T^{(r)}$, each of which has the same distribution as does the test statistic T when H_0 is true. Hence, by the law of large numbers, the proportion of the T_i that are as large as t will be very nearly equal to the probability that T is as large as t when H_0 is true—that is,

$$\frac{\text{number } l : T^{(l)} \geq t}{r} \approx P_{H_0}\{T \geq t\}$$

[1] Optional section.

In fact, by letting r be large, the foregoing can be considered to be, with high probability, almost an equality. Hence, if that proportion is less than or equal to α, then the p-value, equal to the probability of observing a T as large as t when H_0 is true, is less than α and so H_0 should be rejected.

Remarks

(a) To utilize the foregoing simulation approach to determine whether or not to accept H_0 when T is observed, we need to specify how one can simulate, or generate, a random variable Y such that $P\{Y = i\} = p_i, i = 1, \ldots, k$. One way is as follows:

Step 1: Generate a random number U.

Step 2: If

$$p_1 + \cdots + p_{i-1} \le U < p_1 + \cdots + p_i$$

set $Y = i$ (where $p_1 + \cdots + p_{i-1} \equiv 0$ when $i = 1$). That is,

$$\begin{aligned} U < p_1 &\Rightarrow Y = 1 \\ p_1 \le U < p_1 + p_2 &\Rightarrow Y = 2 \\ &\vdots \\ p_1 + \cdots + p_{i-1} \le U < p_1 + \cdots + p_i &\Rightarrow Y = i \\ &\vdots \\ p_1 + \cdots + p_{n-1} < U &\Rightarrow Y = n \end{aligned}$$

Since a random number is equivalent to a uniform (0, 1) random variable, we have that

$$P\{a < U < b\} = b - a, \qquad 0 < a < b < 1$$

and so

$$P\{Y = i\} = P\{p_1 + \cdots + p_{i-1} < U < p_1 + \cdots + p_i\} = p_i$$

(b) A significant question that remains is how many simulation runs are necessary. It has been shown that the value $r = 100$ is usually sufficient at the conventional 5 percent level of significance.[2]

Example 11.2.c. Let us reconsider the problem presented in Example 11.2.b. A simulation study yielded the result

$$P_{H_0}\{T \le 9.52381\} = .95$$

[2]See Hope, A., "A Simplified Monte Carlo Significance Test Procedure," *J. of Royal Statist. Soc.*, B. 30, 582–598, 1968.